MIL-C-23261(CG)
10 April 1962

MILITARY SPECIFICATION

COMPUTER, DITCH HEADING

1. SCOPE

1.1 Scope.- This specification covers ditch heading computers of one type.

2. APPLICABLE DOCUMENTS

2.1 The following military standard, of the issue in effect on date of invitation for bids, forms a part of this specification to the extent specified herein:

MIL-STD-129 Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue (Code CDS), Philadelphia 20, Pennsylvania).

3. REQUIREMENTS

3.1 Materials.- Materials shall be as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.1.1 Metals.- Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.1.2 Plastic.- The vinyl resin used shall be homogeneous, flat, and of a type having minimum water absorption and a minimum tendency to warp.

3.2 Construction.- The computer shall consist of three concentric discs and three indicator arms assembled on a common pivot. The ditch heading indicator arm shall be mounted between the center and back discs. The two divergence indicator arms shall be mounted behind the back disc. When fully assembled, computer shall appear as shown in Figure 1 for the front view, Figure 3 for the back view, and Figure 4 for the side view.

FSC 6605

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3.2.1 Discs.- The discs of the computer shall be fabricated of vinyl copolymer resin. The discs shall be transparent except for printed and screened areas.

3.2.1.1 Disc laminations.- All printing and screening applied to plastic shall be laminated with a transparent plastic sheet of 0.002-inch minimum thickness and be sealed against the entrance of moisture between the laminations. To insure proper sealing, no screening or printing shall be closer than one-eighth-inch from the disc edges.

3.2.1.2 Discs - Alternative Construction.- The discs may be fabricated of paper, laminated between equal thicknesses of transparent vinyl. The diameter of the plastic laminations shall be greater than that of the paper discs to allow a sufficient rim (at least one-eighth-inch) for heat sealing. All perforations or cut-outs in the discs shall be sealed against entrance of moisture between the laminations.

3.2.1.3 Diameter of discs.- The diameter of the working areas of the discs shall be as follows ± 1/32-inch:

a. Back disc (ditch heading and wind scale) - 8-inches

b. Center disc (major swell) - 6-3/8-inches

c. Top disc (minor swell) - 4-1/8-inches

The total diameter of each disc shall not exceed the diameter of the working area by more than 3/8-inch.

3.2.2 Indicator arms.- The ditch heading indicator arm and the two divergence indicator arms shall be of design conforming to Figures 1, 3, and 4. Width of the shafts of the arms shall not exceed 1/4-inch. The indicator arms shall be of metal conforming to 3.1.1. Unless otherwise specified, the indicator arms shall be coated with an orange finish which shall be smooth and highly resistant to chipping or cracking.

3.2.3 Thickness of discs and indicator arms.- All discs and indicator arms shall be of 0.040 ± 0.004 inch thickness.

3.2.4 Assembly.- A flat plastic washer of 1/64-inch nominal thickness and 3/4 to 1-inch diameter shall be inserted between the top (minor swell) and center (major swell) discs. A metal eyelet and spring washer shall be used in assembling the discs, arms, and washer concentrically, in accordance with Figure 4. The eyelet and spring washer shall be of adequate strength to resist the normal forces resulting from handling and manipulating the computer.

3.2.5 Movement.- The discs and arms shall move easily and smoothly and yet have sufficient friction to retain their positions in normal use.

3.3 Markings.- All markings, including letters and numbers, shall be white on black, black on white, or black on colors and shall be sharply defined and legible, in accordance with Figures 1 and 2.

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3.3.1 Background behind markings.- The background behind the markings on the back disc (ditch heading and wind scale) shall be green. The background behind the numbers on the rim of the center disc (major swell) shall be black. The "best" and "fair" fields on the center (major swell) and top (minor swell) discs shall be pale green, slightly intensified on the axis of the word "fair" and gradually intensified to a medium shade of green on the axis of the words "best". The "poor" fields on the major and minor swell discs shall be pale red, gradually intensified to a medium red along the axis of the indicator arrows.

3.4 Instructions for use of computer.- Instructions for use of the computer shall be printed on the back of the laminated sheet of the back disc, as shown on Figure 2. In order not to obscure the printed data, the divergence indicator arms are not shown on Figure 2. The instructions shall include the following information, as shown in Figure 2:

a. A table showing the maximum allowable heading divergence in relation to true wind velocity.

b. A graph showing the theoretical relationship between wave length, velocity, and period in deep water.

c. Data for a sample problem, with a step by step solution and a reproduction of the face of the computer as it looks at the solution of the problem.

3.5 Finish.- All surfaces of the discs and indicator arms shall have a smooth finish.

3.6 Identification.- The information below shall be permanently and legibly marked on the face of each computer:

COMPUTER, DITCH HEADING

Contract or Order No.

Stock No.

Manufacturer's name or trade-mark

U. S. Property

3.7 Workmanship.- The computer, including all parts, shall be free of burrs, sharp edges or any other evidence of poor workmanship.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample.- Unless otherwise specified, for each contract or order, two computers shall be manufactured and submitted to the Government inspector who shall examine the sample to verify strict conformance with drawings and specifications prior to production tests at an activity designated in the contract or by the procuring activity. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.3 Lot size.- As applied to Government inspection of units or product the term "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.4 Classification of inspection

(a) Acceptance Inspection

(b) Preproduction Inspection

4.5 Acceptance inspection.- These inspections are to be accomplished on each computer being submitted for acceptance under contract.

4.5.1 Examination of product.- Each computer shall be inspected to determine conformance with this specification with respect to material, assembly, design, markings, and workmanship.

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4.5.2 Scale error.- The back (ditch heading and wind scale) disc shall be mounted on a suitable turntable. Scale error shall be determined by aligning the zero heading of the computer disc with the zero index of the turntable and then rotating disc 360 degrees,through increments of 5 degrees, the increments being read on the turntable scale. The error shall be determined by the amount the computer disc differs in reading from that of the turntable scale reading. Scale error shall not exceed one degree. This test shall be repeated with the center (major swell) disc.

4.5.3 Disc layouts.- The accuracy with which the designs are laid out shall be determined by measuring the diameter of one circle on each disc of the computer. The diameter of the circle as measured from point to point around the circle shall not vary more than 1/64 inch. This measurement shall be made prior to assembly of the computer.

4.6 Preproduction inspection.- Preproduction inspection shall consist of all the inspections contained in this specification.

4.7 Rejections.- Percent of units defective above 2.5 percent shall be cause for rejection of the entire lot.

4.7.1 Defective units.- Units found defective in the course of inspection shall be rejected, whether or not the lot as a whole is acceptable.

4.7.2 Resubmitted lots.- Lots found unacceptable shall not be resubmitted for reinspection unless all defective units are removed or all defects corrected by the supplier. The resubmitted lots shall be reinspected using either normal or tightened inspection for all types or classes of defects or merely reinspected for the particular types or classes of defects at the discretion of the Government.

5. PREPARATION FOR DELIVERY

5.1 Unit package.- Each computer shall be enclosed in a kraft envelope having a bursting strength (Mullen) of 56 pounds per square inch, end opening, with gummed flap. Size of envelope shall be 9-inches by 9-inches or sufficient to close flap with computer in the envelope. Other packaging shall be in accordance with the manufacturer's commercial practice.

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5.2 Packing.- Unless otherwise specified, interior packages shall be packed in substantial commercial shipping containers constructed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of the Uniform Freight Classification.

5.3 Marking.- In addition to markings required by contract or order, unit package, intermediate package, and outer container shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.- The ditch heading computer is intended for use in advising pilots of ditching aircraft of the best heading on which to put down their planes in the open sea.

6.2 Ordering data.- Procurement documents should stipulate the following:

(a) Title, number, and date of this specification.

(b) Federal stock number.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity:
U. S. Coast Guard

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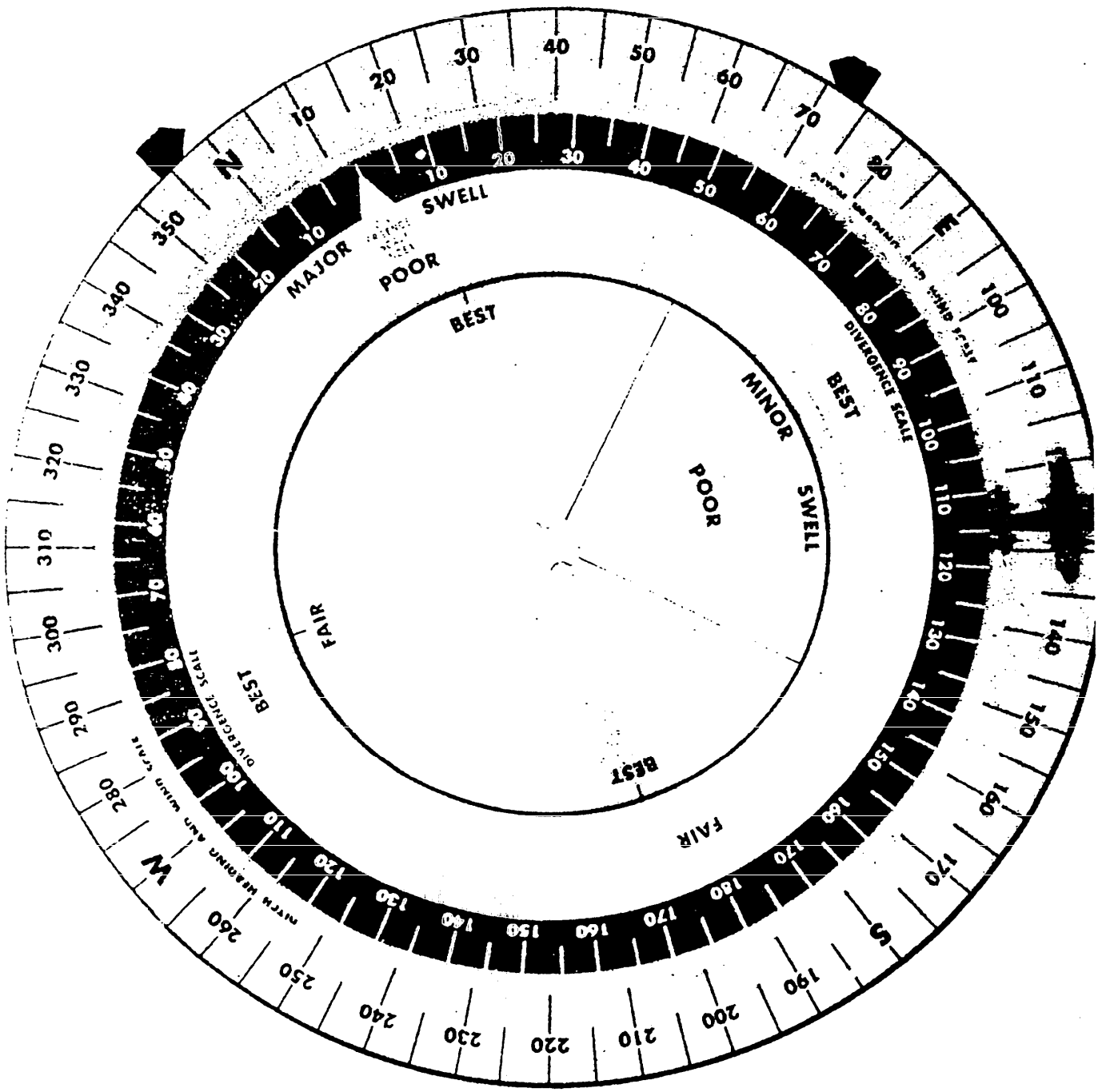


FIGURE 1

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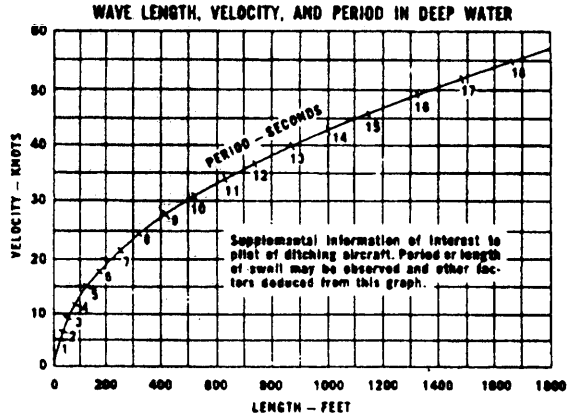
INSTRUCTIONS FOR USE OF DITCH HEADING COMPUTER

MAXIMUM ALLOWABLE HEADING DIVERGENCE

Divergence Table

True Wind Velocity (knots)	Heading Divergence (degrees)
0	180
5	180
10	125
15	125
20	100
25	90
30	55
35	45
40	40
45	35
50	30

THEORETICAL RELATIONSHIP BETWEEN WAVE LENGTH, VELOCITY, AND PERIOD IN DEEP WATER



SAMPLE PROBLEM

Wind 300° 10 Kts
Major Swell 315°
Minor Swell 270°

SOLUTION

1. Use wind velocity (10 Kts) under "True Wind Velocity" scale.
2. Note opposite under "Heading Divergence"— 125°.
3. Set divergence indicator index arrow at 300°.
4. Set divergence indicator arms 125° each side of index arrow. (This locates divergence arms at 85° & 175° respectively on wind scale.)
5. Re-set major swell index arrow at 315°.
6. Set minor swell index arrow at 270°.
7. Move ditch heading indicator to combination of darkest green areas on major and minor swell discs giving preference to major swell disc.

Result—read ditch heading of 40° on ditch heading scale. See illustration at right.



NOTE
Always give preference to MAJOR swell scale unless major and minor swells approximate each other in height, then equal weight shall be given each.

FIGURE 2

(Showing data on back of back disc with divergence indicator arms detached)

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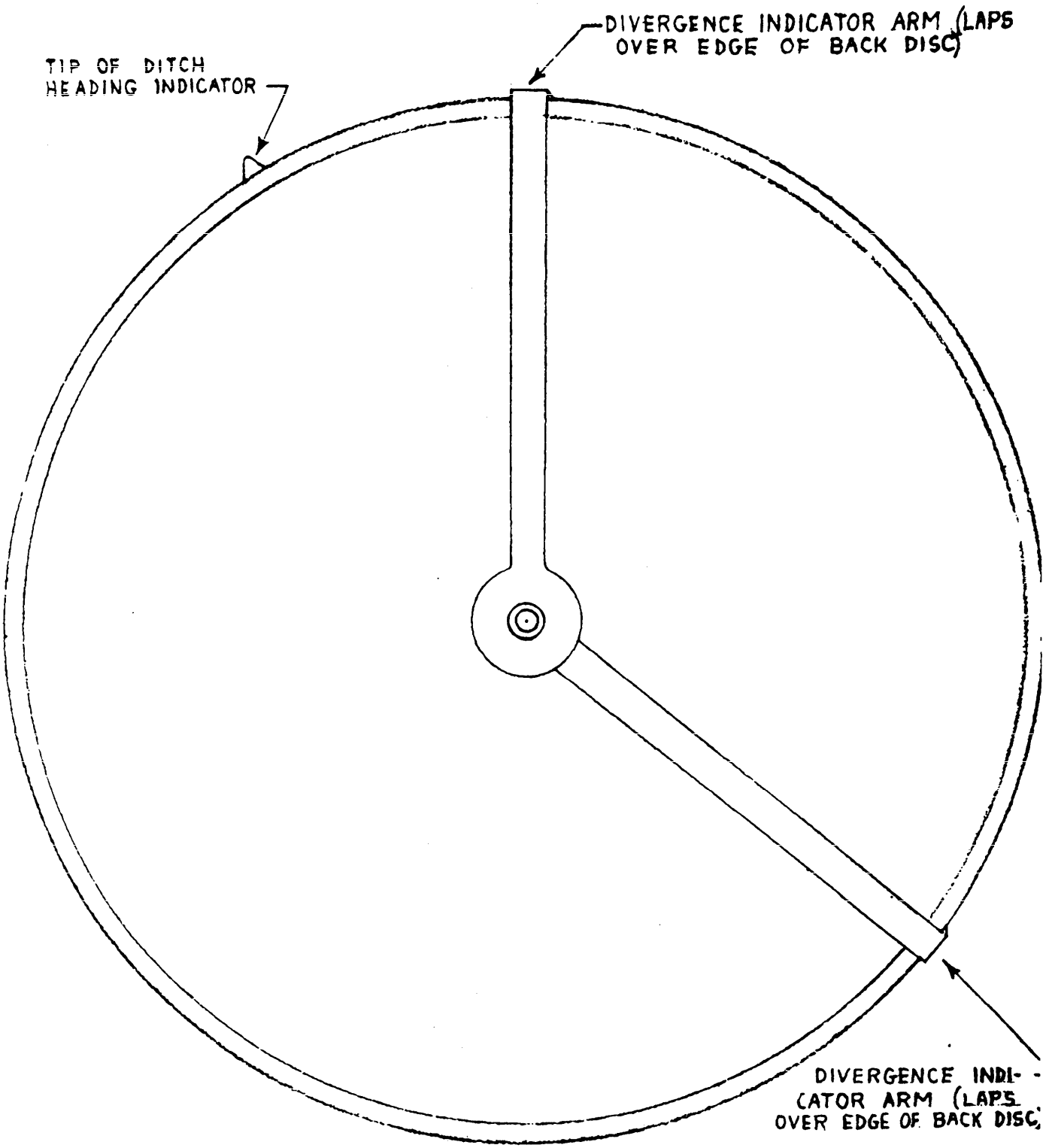


FIGURE 3

(Showing assembly at back of computer with divergence indicator arms attached)

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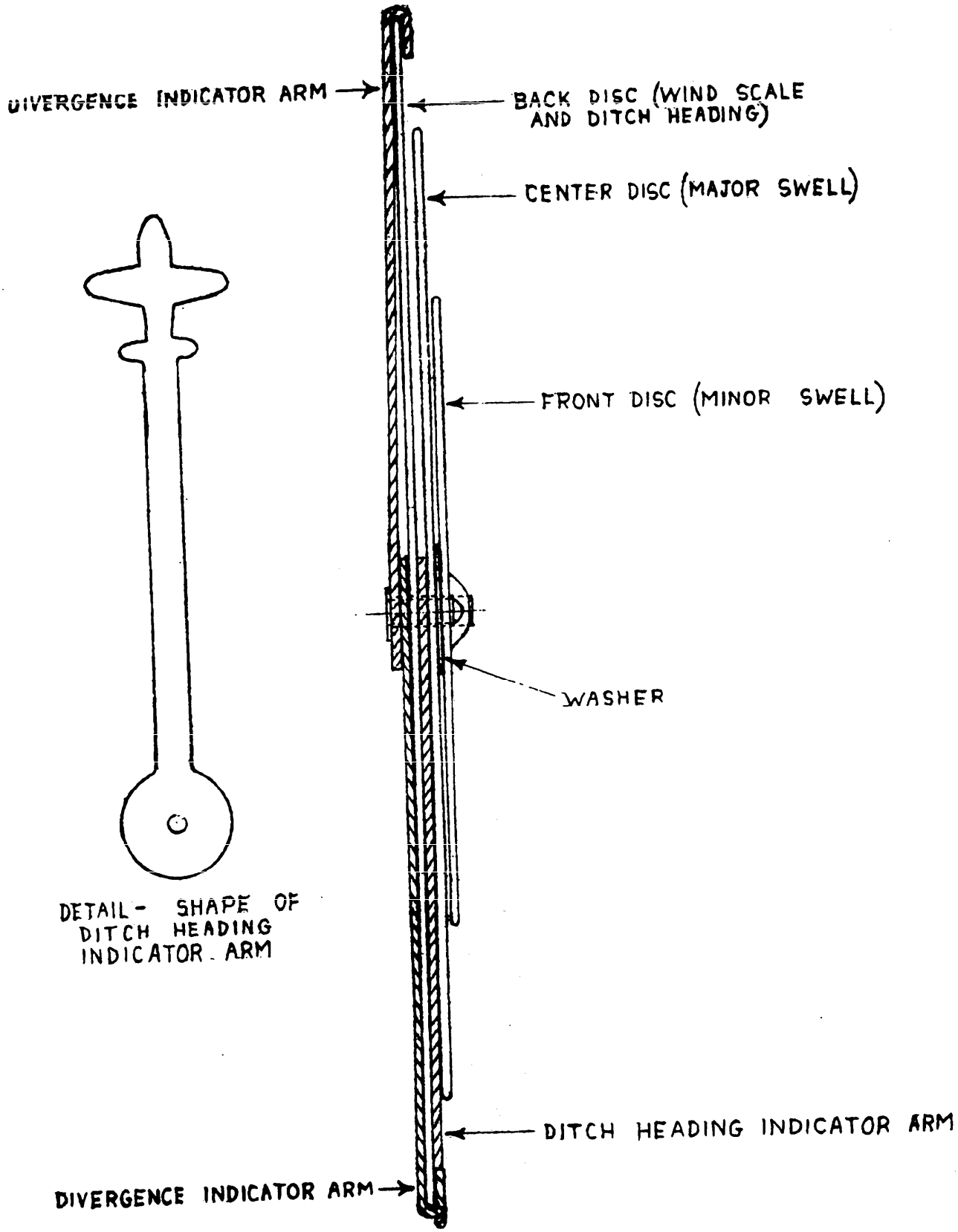


FIGURE 4

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